

NeuroLens: An Interactive Tool for Efficient EEG Annotation

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ABSTRACT

Epilepsy is a chronic neurological disorder characterized by recurrent seizures, and despite decades of research, it remains difficult to manage and fully understand. Although many studies have explored automated seizure detection using EEG data, a significant portion of this work relies either on a limited number of publicly available datasets or on costly proprietary datasets obtained through clinical collaborations. This scarcity and expense of high-quality annotated data pose a major barrier to scalable and robust model development. To address this challenge, we present NeuroLens, an interactive visual analytics system designed to improve the efficiency and reliability of EEG annotation. NeuroLens integrates active learning within a human-in-the-loop framework, enabling clinical experts to focus on EEG samples where the model exhibits high uncertainty. By prioritizing the most uncertain instances, the system reduces annotation effort and cost while maintaining data quality and clinical relevance. For interpretability, the system employs a Logistic Regression model to classify EEG segments into seizure and non-seizure categories. The dashboard is organized into three primary components: (1) an Overview Panel with embedding space visualization and uncertainty histogram for global insight, (2) an Inspection Panel with EEG annotation and feature importance views for detailed analysis, and (3) an Evaluation Panel with learning curves and prediction flow visualizations to track model performance across iterations. By combining interpretable modeling, active learning, and interactive visualization, NeuroLens demonstrates an effective approach to reducing annotation burden while fostering trust and usability in medical machine learning workflows.

1 INTRODUCTION

Medical data annotation, is inherently expensive, time-consuming, and requires significant domain expertise. High-quality labeled datasets are hard to obtain because of strict regulations, privacy concerns, and the need for expert clinicians to interpret complex data. This scarcity of annotated data presents a major bottleneck in developing reliable machine learning models for clinical applications such as epileptic seizure detection.

EEG analysis plays a critical role in diagnosing neurological disorders like epilepsy. However, annotating EEG recordings, especially identifying seizure and non-seizure patterns, is a cognitively demanding task that requires substantial clinical expertise and time investment. In practical settings, manually labeling large volumes of EEG data is not scalable, limiting the availability of annotated datasets for training robust models.

To address this challenge, NeuroLens is proposed as a visual analytics system designed to improve the efficiency and effectiveness of EEG annotation. The approach leverages an active learning framework, where the system iteratively selects the most uncertain samples for expert annotation. An initially unlabeled dataset is assumed, where seizure and non-seizure labels are unknown, and the objective is to efficiently generate high-quality annotations with minimal expert effort. By prioritizing uncertain samples, NeuroLens

reduces redundant labeling and focuses expert attention where it is most impactful.

The system is built on EEG data from the University of Bonn dataset, consisting of 500 EEG segments. Given the time-series nature of the signals, applies feature engineering to extract both time-domain and frequency-domain characteristics for classification. An interpretable logistic regression model is integrated within the active learning loop to classify EEG segments into seizure and non-seizure categories, balancing interpretability with performance.

Seizure detection often relies on fully labeled datasets processed through automated pipelines [5] [1]. While some studies explore automated detection using handcrafted features and classifiers such as SVMs, fewer address the challenge of reducing annotation effort through human-in-the-loop systems and interactive visualization. Furthermore, despite reviews highlighting the growing complexity of EEG analysis and the need for scalable solutions, the integration of active learning with interpretable models and visual analytics remains relatively underexplored in this domain [1].

This work bridges this gap by combining active learning, interpretable modeling, and interactive visualization into a unified system. The key contributions of this report are:

- An interactive visual analytics system (NeuroLens) is designed for efficient EEG annotation in a human-in-the-loop setting.
- Active learning is integrated to prioritize uncertain samples, significantly reducing annotation effort.
- Interpretable modeling supports clinical reasoning and builds user trust.
- Visualization is demonstrated to support decision-making, improve annotation strategies, and enable better understanding of model behavior.

Overall, NeuroLens provides a scalable and practical approach to mitigate the challenges of limited labeled medical data, enabling more efficient and informed EEG analysis workflows.

2 DATA, MODEL, USER AND TASK DESCRIPTION

2.1 Dataset and Pre-processing

The dataset used in this study is the University of Bonn EEG Dataset [2], a widely recognized benchmark for epileptic seizure detection. It comprises 500 single-channel EEG segments, organized into five subsets (A–E) with 100 segments each, ensuring a balanced class distribution. Each segment represents approximately 23.6 seconds of brain activity sampled at 173.61 Hz, resulting in 4,096 continuous amplitude measurements stored in ASCII format. The subsets correspond to clinically meaningful conditions: Set A (healthy, eyes open), Set B (healthy, eyes closed), Set C (interictal activity from the epileptogenic zone), Set D (interictal activity from the opposite hemisphere), and Set E (ictal activity during seizures). While the dataset supports multi-class classification, it is reformulated in this work as a binary problem, where Set E is labeled as seizure and Sets A–D as non-seizure.

To enable efficient and interpretable learning, the raw EEG signals were preprocessed and transformed into structured features. A low-pass filter with a cutoff at 40 Hz was applied to remove high-frequency noise while preserving relevant brain activity (as seen

in Figure 1). For each segment, time-domain features [3] such as mean, standard deviation, root mean square (RMS), peak-to-peak amplitude, skewness, and kurtosis were extracted to capture statistical characteristics of the signal. Additionally, frequency-domain features [4] were computed using Welch’s Power Spectral Density method, from which band-specific powers (delta, theta, alpha, and beta) were derived.

2.2 Model(s)

The system employs a Logistic Regression classifier for binary classification of EEG segments into seizure and non-seizure classes. The model is implemented using a scikit-learn pipeline consisting of feature standardization (StandardScaler) followed by Logistic Regression with the saga solver. Hyperparameters are optimized using GridSearchCV over a search space of regularization strength ($C \in \{1, 0.1, 0.01\}$) and penalty types (L1 and L2), with GroupK-Fold (5-fold) cross-validation to ensure subject-wise separation and prevent data leakage. The dataset of 500 EEG segments is split using GroupShuffleSplit with an 80:20 ratio, resulting in approximately 400 training samples and 100 testing samples, while maintaining subject independence. To align with clinical priorities, a custom classification threshold is used. Instead of the default 0.5, a segment is classified as a seizure if the

$$\text{predicted_probability} \geq 0.30$$

, prioritizing sensitivity and reducing false negatives. Model performance is evaluated using accuracy, sensitivity, and specificity, along with confusion matrix analysis.

The model is embedded within an active learning framework to reduce annotation cost and improve labeling efficiency. Initially, 10% of the training data (~ 40 samples) is randomly selected and labeled to train the initial model, while the remaining samples form an unlabeled pool. In each iteration, the model computes prediction probabilities for unlabeled samples and selects a batch of 10 samples (batch size = 10) based on uncertainty. The uncertainty of a sample x is defined as:

$$u(x) = 1 - \max_c P(y = c | x)$$

i.e., one minus the model’s confidence in its most probable class. A confidence threshold (default = 0.70) is used to divide samples into two groups: (1) low-confidence samples requiring expert (oracle) annotation, and (2) high-confidence samples that are automatically accepted by the model. Only uncertain samples are presented to the user through an annotation queue, ensuring efficient use of expert effort.

After each annotation phase (up to 10 newly labeled samples per round), the model is retrained using the expanded labeled dataset. This iterative process continues until a hybrid stopping criterion is satisfied: (i) no samples fall below the confidence threshold (i.e., no further expert annotation is required), and (ii) model predictions stabilize across consecutive rounds for 2 consecutive rounds. This is measured by as:

$$r_{flip} = \frac{\#prediction_flips}{totalsamples} < 0.01.$$

This ensures that the model has converged and additional annotations would yield minimal improvement. Overall, this framework tightly integrates classification and active learning, where the model not only performs prediction but also actively guides the data annotation process.

2.3 Users

The intended users for this system are domain experts who work with EEG recordings for epilepsy diagnosis. It is particularly intended

for professionals who have moderate technical or machine learning expertise but have extensive knowledge in their clinical domain. The primary analytical need of these users is to quickly distinguish between normal and abnormal brain activity, focus on clinically significant patterns (e.g., spikes, frequency changes), and validate model predictions with confidence. Additionally, they require tools that help them understand why a model makes a particular decision, so that it aligns with clinical reasoning and builds trust in automated assistance. The NeuroLens dashboard directly addresses these needs as it prioritizes and presents only those EEG samples where the model is uncertain, significantly reducing the annotation burden.

2.4 Analytical Tasks

Task 1: Optimize Annotation Strategy to Maximize Learning Efficiency: This task focuses on how visual analytics can guide experts to annotate effectively within the active learning loop. Rather than annotating randomly, the system enables users to identify and prioritize the most uncertain samples (or informative samples) that are likely to yield the greatest improvement in model performance. By interactively adjusting confidence thresholds and exploring uncertainty distributions, experts can balance annotation cost against model gains, making the learning process more efficient and targeted. Key questions the task addresses are as follows:

- Which samples are truly useful for improving the model, as opposed to being noise or outliers?
- How does adjusting the confidence threshold influence the trade-off between annotation effort and performance improvement?

Task 2: Evaluate and Improve Model Reasoning: This task enables experts to move beyond accepting model predictions and instead critically assess the reasoning behind them. By linking feature importance with EEG signal patterns, users can validate whether the model’s decisions align with clinical knowledge. This process helps identify incorrect assumptions, biases, or spurious correlations, allowing experts to iteratively refine trust in the system and guide improvements in feature representation or model behavior. Key questions the task addresses are as follows:

- Do the model’s feature attributions correspond to clinically meaningful EEG patterns, or do they reveal misleading correlations?
- In cases of incorrect or uncertain predictions, what do the feature contributions indicate?

Task 3: Identify and Reduce Ambiguity in Classification: This task leverages embedding visualizations to provide a global view of the dataset and model behavior. By exploring how seizure and non-seizure samples are distributed in the embedding space, experts can identify regions of overlap and ambiguity where the model struggles. These insights enable targeted interventions, such as selective annotation or threshold adjustment, to improve class separability and refine the decision boundary. Key questions the task addresses are as follows:

- Where do seizure and non-seizure samples overlap in the embedding space, and what does this indicate about data ambiguity?
- How can targeted annotation help improve the performance of the model?

Task 4: Monitor Model Learning Dynamics and Determine Convergence: This task focuses on evaluating how the model evolves over successive active learning rounds and determining when further annotation is no longer beneficial. By analyzing performance trends

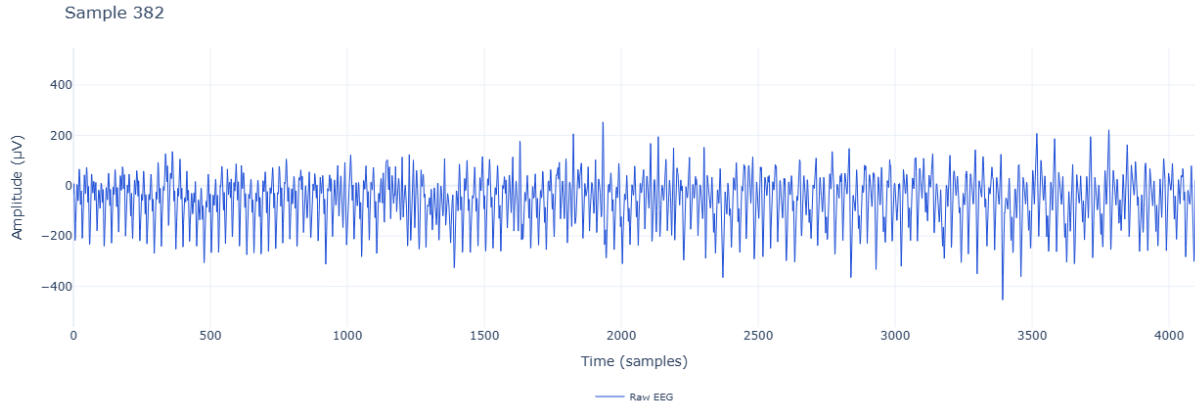


Figure 1: EEG sample of a patient.

and prediction transitions, users can assess whether improvements are meaningful or diminishing. Stability indicators, such as reduced prediction changes, help determine convergence, ensuring that annotation efforts are stopped at the optimal point. Key questions the task addresses are as follows:

- How do performance metrics and prediction transitions evolve across rounds?
- Do stability indicators suggest that the model has converged, or is further annotation required?

3 VA SOLUTION DESIGN

NeuroLens is designed not merely as a classification tool, but as a visual analytics system that places the clinical expert at the center of the annotation process. The entire system is contained within a single screen, ensuring that all relevant information remains simultaneously visible and eliminating the cognitive overhead of navigating between separate views. The design (as seen in Figure 2) prioritizes global understanding before local inspection so that the user can first develop an intuition about the overall data landscape before committing to individual annotation decisions.

A known challenge in active learning is the cold start problem. A model with no labeled data cannot produce meaningful uncertainty estimates, making it impossible to determine which samples to present first. NeuroLens addresses this as an explicit design decision by pre-labeling a small fraction of the samples, providing the model with sufficient EEG samples to generate an initial uncertainty distribution. This bootstrapped model then predicts labels and computes uncertainty scores for the remaining unlabeled pool, and the system enters its iterative annotation workflow.

The Overview Panel serves as the primary entry point into the system, presenting the UMAP embedding space as the first view the expert encounters, as seen in Figure 3. UMAP is chosen over linear dimensionality reduction techniques such as PCA because it is a local structure-preserving method that prioritizes topological neighborhood relationships over global distance preservation. This is critical here because the goal is not merely to separate the two classes but to expose the regions near the decision boundary where seizure and non-seizure samples are most ambiguous, i.e., the precise regions that active learning needs to target. Since uncertain samples naturally concentrate near the decision boundary, visual overlap in the embedding becomes a direct usability concern. DGrid-inspired layout (that uses a force-based spreading mechanism) is therefore applied to spatially separate overlapping points while preserving

their relative distribution, ensuring that individual uncertain samples remain selectable and visually distinguishable even in dense boundary regions.

To further support flexible exploration, the system provides three view modes: labeled, unlabeled, and all samples, allowing experts to selectively focus on different subsets of the data depending on their analytical needs, whether it is inspecting model predictions, reviewing annotated samples, or identifying regions requiring further labeling. Additionally, samples that have been annotated by the expert are visually highlighted with a darker border, enabling quick identification of reviewed data and making the progression of the active learning process more transparent. Uncertainty is encoded as a continuous gradient where deep red indicates the highest uncertainty and lighter beige tones indicate higher confidence (Figure 3). A gradient is chosen over discrete categorical colors because uncertainty is a continuous quantity, and this single-hue gradient allows the expert to immediately read uncertainty intensity across the embedding without a legend lookup.

Alongside the embedding, the uncertainty histogram, Figure 4, provides a complementary global view of how prediction confidence is distributed across the entire dataset. While the embedding shows where uncertain samples sit in the model’s learned space, the histogram shows how many samples fall at each level of uncertainty and what their labeling status is - labeled, unlabeled, or auto-classified. Colors are used as follows: unlabeled samples are shown in blue, initially labeled data in light green, samples requiring expert annotation in red, expert-annotated samples in dark green, and the current batch in dark blue. This mapping is thoughtfully designed to reflect both state and priority, for instance, red highlights high-uncertainty samples that demand immediate attention, while shades of green indicate progressively confirmed labels. This color scheme enables the expert to quickly interpret annotation needs and model confidence across views without additional cognitive load. Together, these two views allow the expert to understand both the spatial and distributional character of uncertainty before any annotation decision is made. The histogram supports both linear and logarithmic scale toggles, a design choice motivated by the skewed nature of uncertainty distributions in active learning. In later rounds, as the model becomes more confident, the vast majority of samples concentrate in low-uncertainty bins, making rare high-uncertainty samples nearly invisible on a linear scale. The log scale toggle exposes this tail, ensuring that the most annotation-critical samples remain perceptually accessible regardless of how the distribution evolves across rounds.

Based on what the expert observes in the Overview Panel, samples are surfaced for annotation through two complementary strategies.

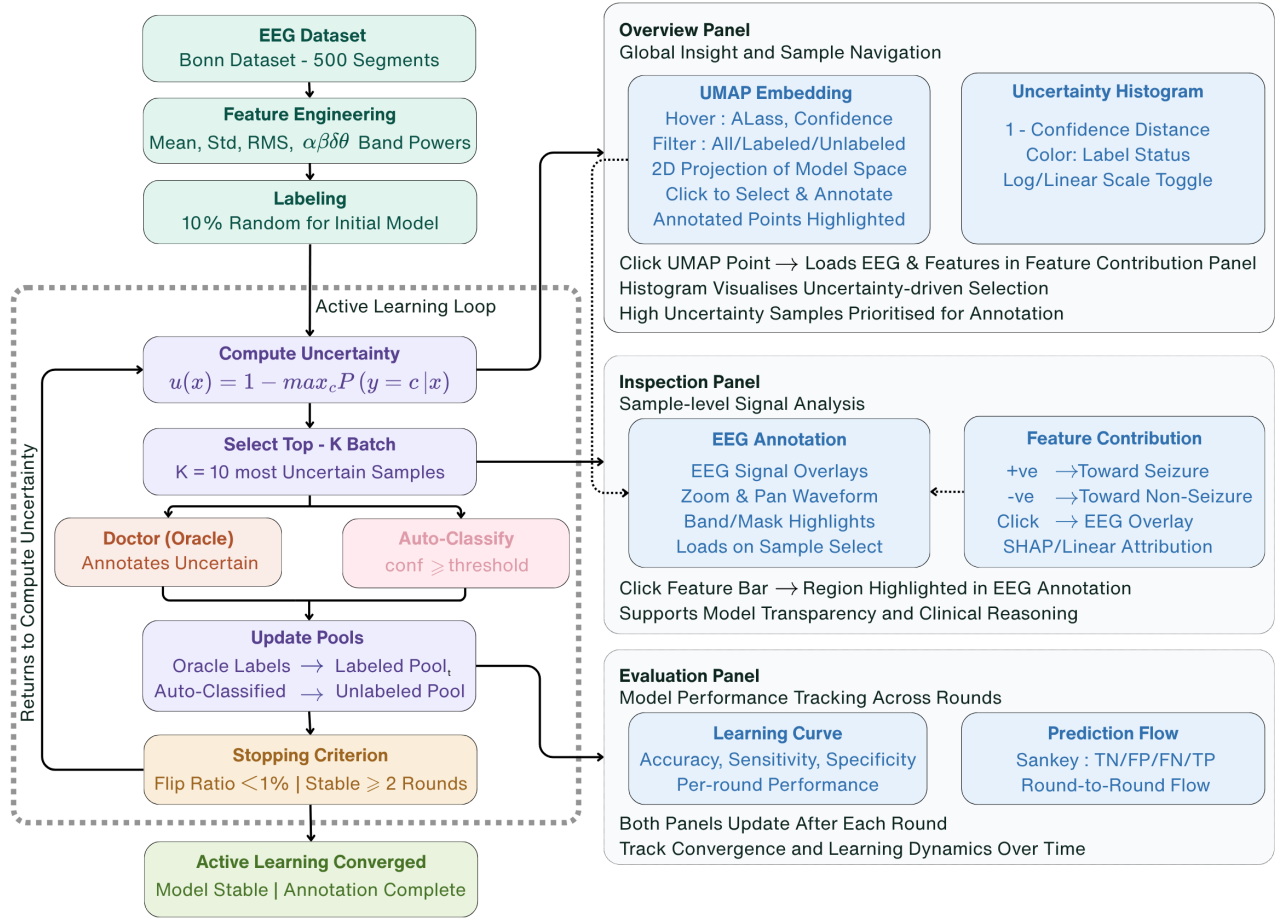


Figure 2: Overall solution design of NeuroLens.

The first is Top-K uncertainty sampling, where the system automatically populates the annotation queue with the most uncertain samples. Those where the model’s confidence is lowest and expert input is most likely to improve the decision boundary. The second is targeted selection, where the expert directly clicks points in the embedding space to annotate samples they identify as ambiguous based on their own clinical reasoning. This dual-strategy design is intentional because Top-K sampling ensures systematic coverage of uncertain regions driven by the model, while targeted selection gives the expert agency to intervene based on patterns visible in the embedding that uncertainty scores alone may not capture. Governing both strategies is the confidence threshold slider in Figure 5, which determines the boundary between samples routed to the expert for annotation and those automatically classified by the model. This control embodies a core design principle, the system aids the expert rather than replacing them, and the expert retains direct control over the degree of automation they trust at any given round. Lowering the threshold increases automation but risks propagating confident misclassifications into the labeled pool; raising it increases annotation burden but improves reliability. Exposing this tradeoff as an interactive control rather than a fixed parameter allows the expert to adapt their annotation strategy as their confidence in the model evolves.

Selected samples are then inspected in the Inspection Panel, which presents the EEG annotation view and the feature contribution panel (Figure 6) as spatially adjacent, tightly coupled views. This layout is deliberate—the feature contributions explain the model’s reasoning

about the signal currently displayed, and physical proximity reinforces this functional relationship without requiring the expert to mentally bridge two distant parts of the interface. Feature importance is computed using linear attribution consistent with Shapley value principles, where each feature’s contribution reflects its marginal impact on the prediction, providing local case-by-case explanations of individual predictions. The feature contribution panel follows the same red-blue color scheme maintained consistently across the dashboard—positive contributions toward seizure are encoded in red and negative contributions toward non-seizure in blue—ensuring color meaning requires no relearning across views. Selecting a feature bar highlights the corresponding region directly in the EEG signal, allowing the expert to simultaneously view all contributing features overlaid on the same signal and verify whether the model is attending to clinically meaningful characteristics. The EEG view supports interactive zooming via rectangular selection, enabling users to focus on specific regions of the signal for closer inspection when needed. As samples are annotated, they are marked with a dark border back in the embedding space, directly linking the Inspection Panel to the global view and allowing the expert to track which regions of the embedding have been labeled and reason about what the model has and has not been exposed to during training.

After each annotation round, the model is retrained on the expanded labeled dataset. In subsequent rounds, samples exceeding the confidence threshold are automatically classified by the model, progressively reducing manual annotation burden as the model matures and becomes more confident. This reduction is directly observable

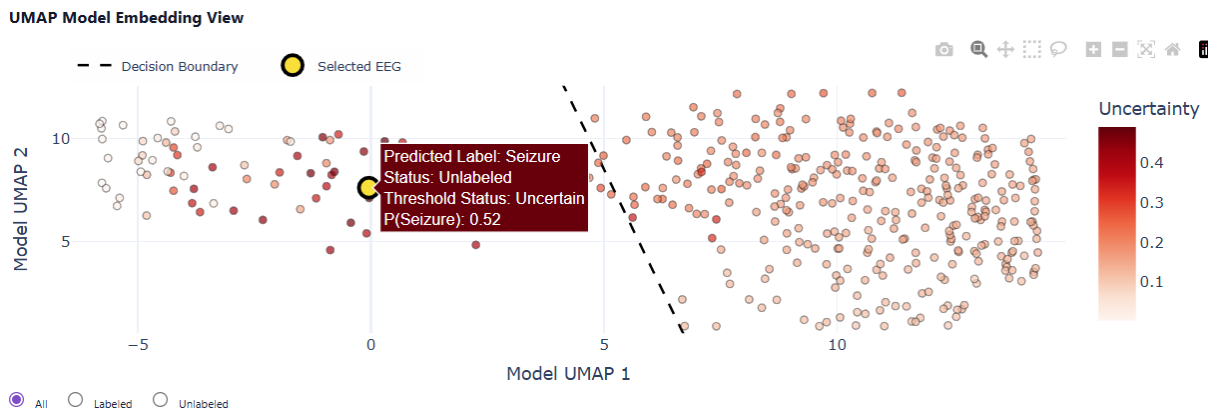


Figure 3: Embedding space using UMAP.

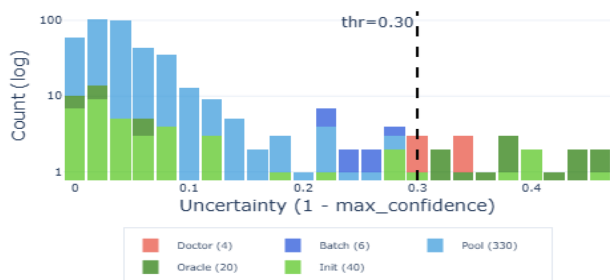


Figure 4: Uncertainty panel that shows distribution of data.

through the information displayed above the uncertainty histogram, which tracks how many of the top-K uncertain samples in each batch of 10 actually require expert intervention—a count that visibly decreases across rounds as the model grows more certain. The expert continues to receive only the samples that genuinely require human judgment, ensuring that annotation effort remains focused where it is most impactful.

The Evaluation Panel, comprising the learning curve and the Sankey diagram (Figure 7), supports monitoring of how the model evolves across rounds. The learning curve (Figure 8) tracks accuracy, sensitivity, and specificity across active learning rounds, allowing the expert to observe how performance responds as increasingly new samples are added to the training set. Sensitivity is tracked alongside accuracy because in seizure detection, a missed seizure carries a significantly higher clinical cost than a false alarm. Aggregate accuracy alone would obscure whether the model is genuinely improving on clinically critical cases or simply maintaining stable overall performance. The Sankey diagram complements this by visualizing how samples transition after model predictions across rounds. True positives are encoded in deep red, true negatives in blue, false positives in green, and false negatives in orange, giving each prediction outcome a distinct and immediately distinguishable identity. Straight flows indicate correct predictions while cross flows represent errors. Flow widths are proportional to sample counts, making the magnitude of each category directly readable. This conveys something a metrics table cannot: not just whether overall performance improved, but whether the right kind of improvement occurred, specifically whether the orange cross flows representing missed seizures are shrinking and the deep red straight flows representing correct seizure detections are growing across rounds, which is the clinically meaningful direction of progress.

4 RESULTS AND EVALUATION

Optimizing Annotation Strategy: Active learning significantly reduced annotation requirements, with strong model performance achieved after labeling only 10–15% (as seen in Figure 9) of the total dataset (approximately 50–75 samples out of the total 500 EEG samples). As training progressed, the number of samples requiring expert intervention decreased, while automatic classification increased, demonstrating improved model confidence. The confidence threshold (default 0.70) played a key role in balancing automation and reliability; lower thresholds increased automation but risked misclassification, whereas higher thresholds increased annotation effort but improved reliability. Visual feedback allowed users to effectively manage this trade-off. Additionally, hybrid labeling using targeted annotation of points likely to flip classes (i.e. point(s) near decision boundary) combined with uncertain samples (depicted in darker colors) led to faster convergence and more stable learning compared to random sampling. The uncertainty histogram (as seen in Figure 4) also showed that annotation requirements reduced over time, validating that expert involvement becomes less necessary as the model matures.

Evaluating and Improving Model Reasoning: The feature importance panel, combined with EEG signal visualization, provided insight into model reasoning. Early in training (as seen in Figure 10), the model relied on a broad set of clinically relevant features and disagreements were observed in the feature contribution panel, i.e. in frequency-based and time domain features. As active learning progressed, the model refined its reasoning and increasingly relied on a smaller subset of highly discriminative features, including RMS, Kurtosis, Theta Power, Mean, and Std Dev, for its decision (as seen in Figure 6). These align well with known seizure indicators (like signals having high amplitude and very high variability), confirming consistency with expert knowledge (as seen in Figure 11–12).

Analyzing Data and Ambiguity: Some regions exhibited clear clustering while others showed overlap between seizure and non-seizure classes. Uncertain samples were initially scattered and were far from the decision boundary (as seen in Figure 13). As learning progressed, ambiguous regions became more structured, although some overlap persisted, reflecting intrinsic EEG ambiguity rather than model failure. The embedding space (as seen in Figure 14) revealed that uncertain samples were consistently concentrated on/near the decision boundary in the subsequent rounds. Importantly, samples that required expert intervention were also observed to predominantly belong to the seizure class, confirming the hypothesis that seizure detection is inherently more complex and ambiguous.

Monitoring Learning Dynamics and Convergence: The evaluation panel, including learning curves and the Sankey diagram (as seen in Figure 8 and Figure 7), provided detailed insights into model

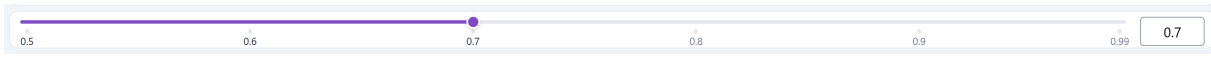


Figure 5: Confidence slider to adjust model sensitivity.

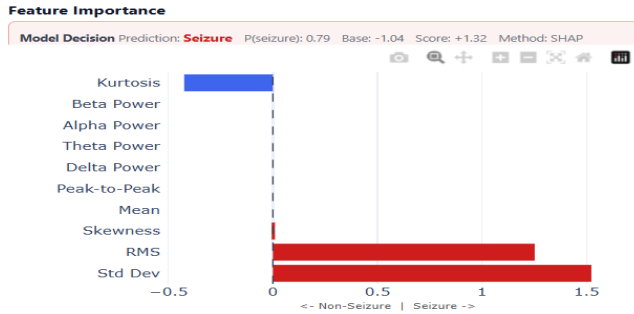


Figure 6: Feature Importance Panel showcasing RMS, Std Dev, Kurtosis as major contributors in later rounds.

behavior. The test accuracy surpasses the training accuracy, indicating the effect of active learning in making the test set relatively easier. At the same time, incorporating highly uncertain samples into the training set reduces training accuracy, as these difficult examples make the learning task more challenging. However, this process improves generalization by exposing the model to edge cases that it would otherwise misclassify. Sensitivity also improves over time and eventually surpasses specificity, indicating better seizure detection. This reflects a clinically meaningful trade-off, with a slight increase in false positives accompanied by a significant reduction in false negatives. The Sankey diagram showed that early rounds produced strong transitions from false negatives to true positives, confirming the effectiveness of targeted annotation, whereas later rounds exhibited minimal transitions, indicating convergence. When annotation was performed randomly, undesirable transitions occurred, such as true positives becoming false negatives and true negatives becoming false positives, highlighting the importance of guided annotation.

5 CONCLUSION AND FUTURE WORK

In this work, we presented NeuroLens, an interactive visual analytics system designed to address the challenge of costly and time-intensive EEG annotation for seizure detection. The goal was to reduce annotation effort while improving model reliability through a human-in-the-loop active learning framework. Our results demonstrate that this objective is largely achieved, as the system is able to reach strong performance using only a small fraction of labeled data, while significantly improving sensitivity for seizure detection.

By integrating interpretable modeling with coordinated visualizations, NeuroLens enables users to understand model behavior, identify uncertain and ambiguous samples, and make informed annotation decisions. This highlights the effectiveness of visual analytics as a tool for interpretation and a mechanism to actively guide and improve the learning process.

However, the problem is not fully solved. The current system relies primarily on uncertainty-based sampling, which may not always capture the most clinically critical errors. Additionally, the system is evaluated on a single dataset and binary classification setup, which limits its ability to generalize to more complex and real-world clinical scenarios. The reliance on simulated oracle annotations also simplifies real-world expert variability.

Future work can extend the system in several directions. A review mechanism can be introduced where experts periodically validate auto-labeled samples to correct confident but incorrect predictions

and improve reliability. The model can also be evaluated on completely unseen EEG data to better assess generalization and robustness in practical clinical scenarios. Additionally, the framework can be expanded beyond binary classification to incorporate seizure types or patient states, making it more aligned with real-world diagnostic workflows.

6 INDIVIDUAL CONTRIBUTION

Both authors jointly contributed to the conceptualization of the system, task design, iterative refinement of the dashboard, report writing, and preparation of the final poster and video.

7 GENERATIVE AI USAGE

Generative AI tools were used in a limited and supportive role during this project, primarily to assist with code structuring suggestions, minor refactoring, and resolving implementation issues; however, all substantive work—including the overall system architecture, active learning workflow design, EEG feature engineering pipeline, custom transformations (such as overlap reduction), dataset splitting strategy, model integration, and the full dashboard interface with its callbacks and visualizations—was independently designed, implemented, and validated by us. External libraries were used strictly as intended (e.g., for numerical computation, signal processing, machine learning models, and visualization primitives), while all higher-level logic, integration across components, and domain-specific reasoning linking model outputs to user-facing insights were developed manually, ensuring that the core intellectual and engineering contributions of the project remain entirely our own.

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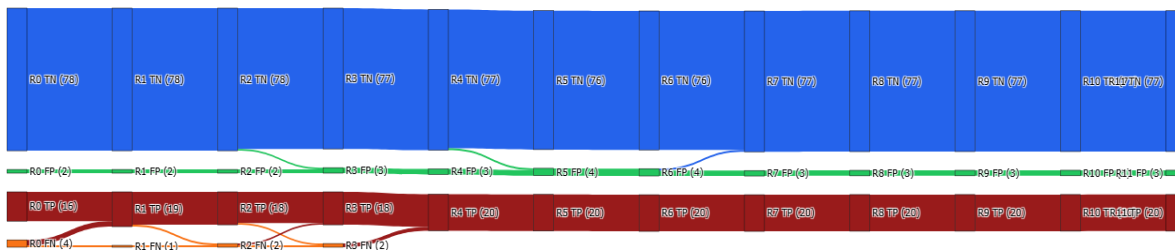


Figure 7: Sankey diagram that shows prediction flow across rounds.

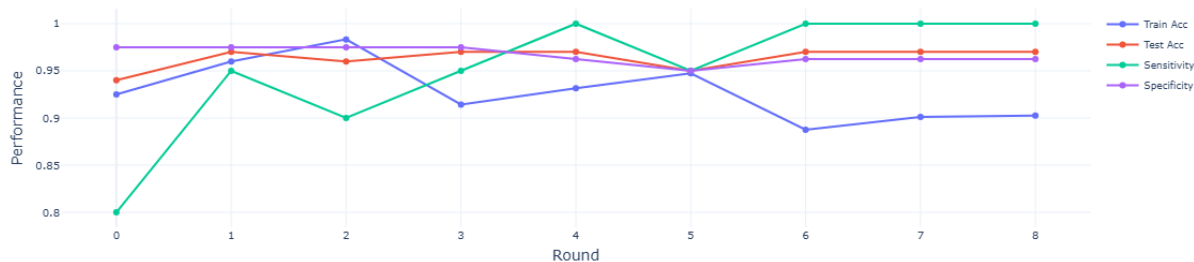


Figure 8: Model learning curve over training iterations.

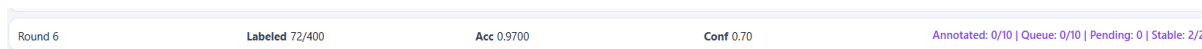


Figure 9: Labeling efficiency achieved with stability across two rounds.

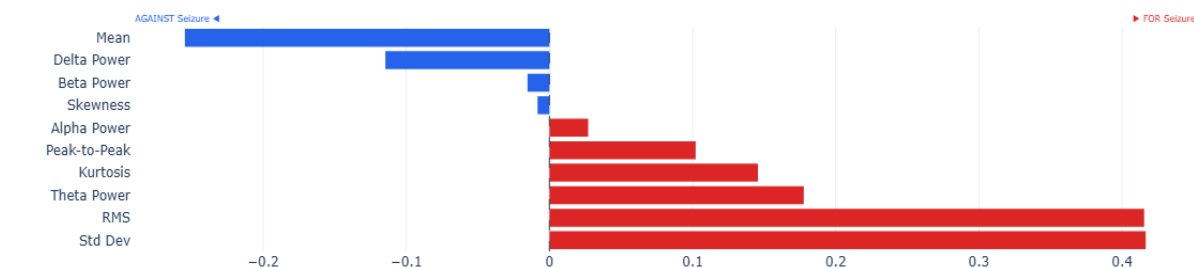


Figure 10: Disagreement in early rounds.

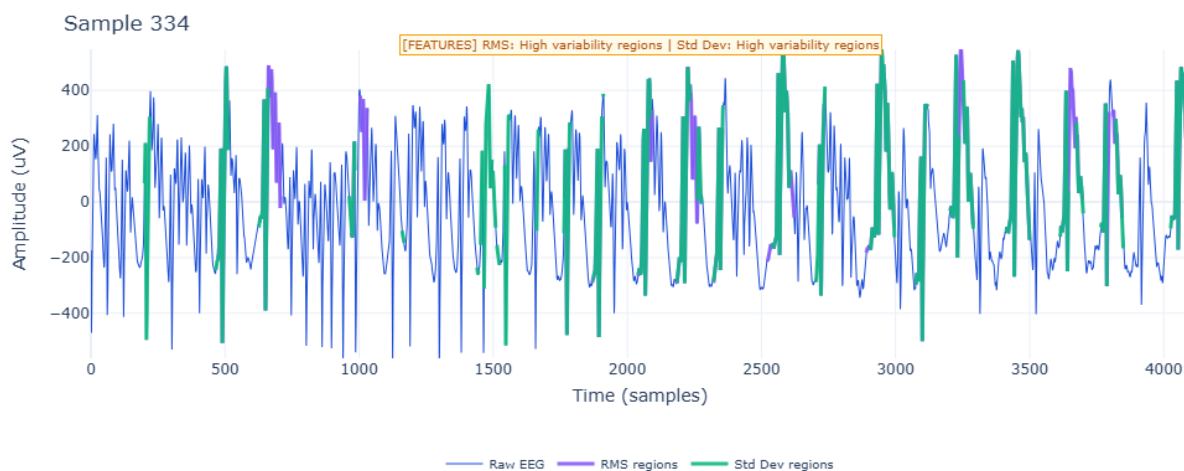


Figure 11: Regions of high variability in an EEG sample belonging to the seizure class.

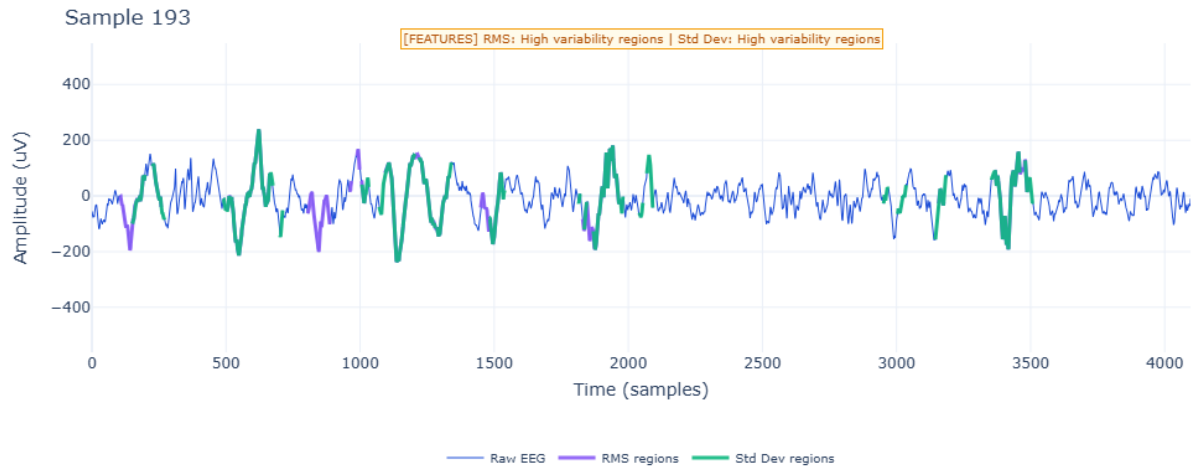


Figure 12: Regions of high variability in an EEG sample belonging to the non-seizure class.

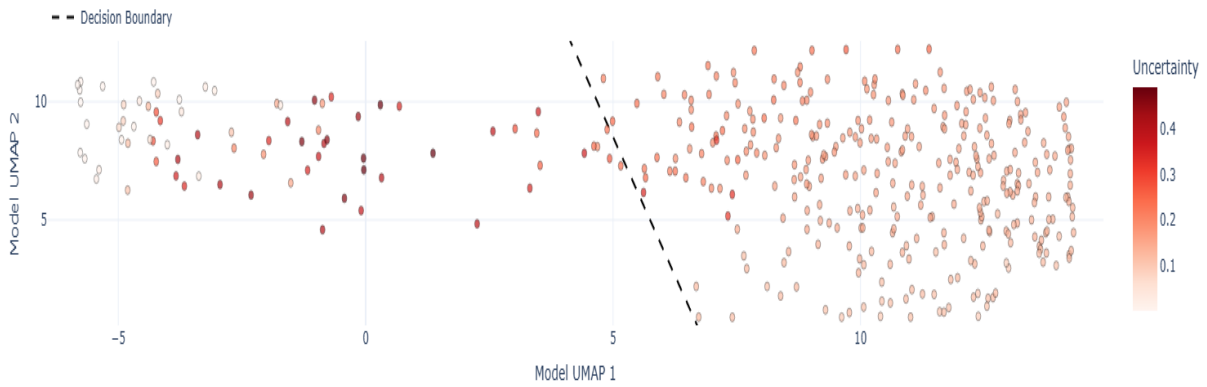


Figure 13: Uncertain points scattered away from the decision boundary in early rounds.

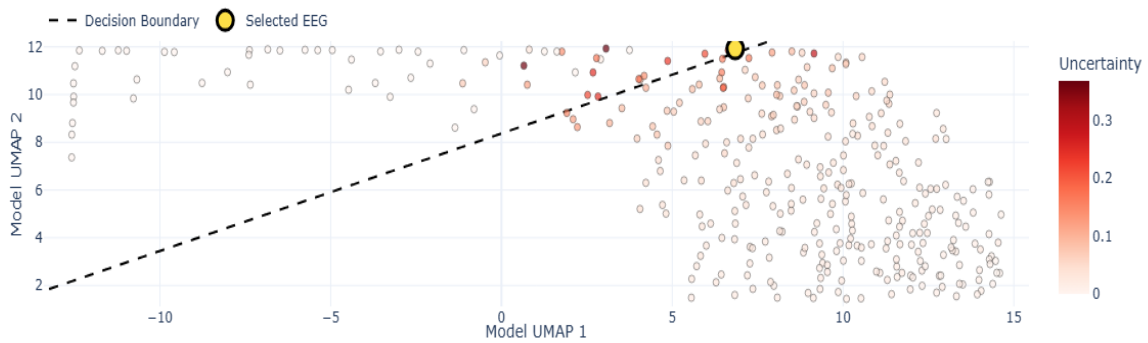


Figure 14: Uncertain regions near the decision boundary.